Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 28: Approved Solutions

The Principle of Mathematical Induction

Covers Chartrand & Zhang, Section 4.1

Status of this set

Ungraded practice. Solutions released with it. Every proof must have a labelled base step and a labelled inductive step.

Part I: How to Practice This Material

The template

1. "We proceed by induction."
2. **Base step:** check $P(1)$ by direct computation.
3. **Inductive step:** "Assume $P(k)$ for a positive integer k. We show $P(k+1)$." Write out $P(k+1)$ explicitly so you know the target.
4. **Use the hypothesis** — and say that you are using it.
5. "By the Principle of Mathematical Induction, $P(n)$ holds for every positive integer n."

For sums: split off the last term

(sum to $k+1$) = (sum to k) + (term $k+1$). The first piece is what the hypothesis describes. Substitute, then simplify toward the target.

Three mistakes to expect

1. **Assuming** $P(k+1)$ and working backward. You may assume only $P(k)$.
2. **Skipping the base step.** Lecture 28 showed a false claim with a valid inductive step.
3. **Never using the hypothesis.** If $P(k)$ is not substituted anywhere, it is not an induction proof.

Part II: Practice Problems

Section A — Sums

1. Prove that $2 + 6 + 10 + \cdots + (4n - 2) = 2n^2$ for every positive integer n.

Approved solution

Proof. We proceed by induction.
Base step. For $n = 1$: left side 2, right side $2(1)^2 = 2$. ✓
Inductive step. Assume $2 + 6 + \cdots + (4k - 2) = 2k^2$ for a positive integer k. We show $2 + 6 + \cdots + (4k + 2) = 2(k + 1)^2$, noting that the $(k + 1)$st term is $4(k + 1) - 2 = 4k + 2$. Then

$$\begin{aligned} 2 + 6 + \cdots + (4k - 2) + (4k + 2) &= 2k^2 + (4k + 2) && \text{by the inductive hypothesis} \\ &= 2(k^2 + 2k + 1) = 2(k + 1)^2. \end{aligned}$$

By the Principle of Mathematical Induction the formula holds for all n. ■

Compute the $(k + 1)$st term before starting. Substituting $k + 1$ into $4n - 2$ gives $4k + 2$, not $4k - 1$ — a slip here makes the rest impossible.

2. Prove that $3 + 7 + 11 + \cdots + (4n - 1) = n(2n + 1)$ for every positive integer n.

Approved solution

Proof. We proceed by induction.
Base step. $n = 1$: left 3, right $1 \cdot 3 = 3$. ✓
Inductive step. Assume $3 + 7 + \cdots + (4k - 1) = k(2k + 1)$. The $(k + 1)$st term is $4k + 3$. Splitting it off and substituting the inductive hypothesis,

$$\begin{aligned} 3 + \cdots + (4k - 1) + (4k + 3) &= k(2k + 1) + (4k + 3) \\ &= 2k^2 + 5k + 3 \\ &= (k + 1)(2k + 3) = (k + 1)\big(2(k + 1) + 1\big). \end{aligned}$$

By induction the formula holds for all n. ■

Factor toward the target. Knowing you need $(k + 1)(2k + 3)$ tells you how to factor $2k^2 + 5k + 3$. Write the target down first and aim at it.

3. Prove that $1 + 5 + 5^2 + \cdots + 5^n = \dfrac{5^{n+1} - 1}{4}$ for every positive integer n.

Approved solution

Proof. We proceed by induction.
Base step. $n = 1$: left $1 + 5 = 6$, right $\frac{25-1}{4} = 6$. ✓
Inductive step. Assume $1 + 5 + \cdots + 5^k = \frac{5^{k+1}-1}{4}$. Splitting off the last term and substituting the

inductive hypothesis,

$$\begin{aligned} 1+5+\cdots+5^k+5^{k+1} &= \frac{5^{k+1}-1}{4}+5^{k+1} \\ &= \frac{5^{k+1}-1+4\cdot 5^{k+1}}{4} \\ &= \frac{5\cdot 5^{k+1}-1}{4} = \frac{5^{k+2}-1}{4}. \end{aligned}$$

By induction the formula holds for all n. ■

Note the sum starts at $5^0 = 1$, so "$n = 1$" means two terms. Read where a sum begins before checking the base step.

Section B — The base step matters

4. Consider the claim: $n^2 + n + 3$ *is even for every positive integer* n.

(a) Show that the inductive step can be carried out.

(b) Evaluate $n^2 + n + 3$ at $n = 1$.

(c) What do (a) and (b) together tell you?

Approved solution

(a) Assume $k^2 + k + 3 = 2\ell$ for some integer ℓ. Then

$$(k+1)^2 + (k+1) + 3 = (k^2 + k + 3) + 2k + 2 = 2\ell + 2(k+1) = 2(\ell + k + 1),$$

which is even. The inductive step works.
(b) $1 + 1 + 3 = 5$, which is odd.
(c) The claim is **false**. The inductive step is valid but there is no true base case to start from. In fact $n^2 + n = n(n+1)$ is always even, so $n^2 + n + 3$ is always *odd* — the hypothesis of the inductive step is never satisfied, and the step is true only vacuously.

This is Lecture 28's warning in a new costume. An inductive step proves an implication, and implications can be vacuously true. The base step is what makes the chain real.

Section C — Writing

5. In five to eight complete sentences, explain why induction proves a statement about *every* positive integer even though it checks only one case directly. Connect your answer to the Week 1 observation that checking cases is not proof.

Approved solution

Model answer. Checking cases can only ever establish finitely many instances, so it cannot settle a claim about all positive integers — which is why $n^2 - n + 41$ misled us in Week 1. Induction does something different: it proves one case directly and then proves a single *universal* implication, that whenever the claim holds for k it also holds for $k + 1$. That implication is not checked case by case;

it is established by an argument about an arbitrary k, so it covers every k at once. Together the base case and the implication force the claim to hold for 1, hence for 2, hence for 3, and so on through every positive integer. So induction does not escape the Week 1 lesson — it respects it, by replacing infinitely many checks with one general argument. The finite work is the base step and the proof of the implication, and the Principle of Mathematical Induction is the axiom that turns those two finite pieces into a conclusion about infinitely many integers.

Full credit requires the point that the inductive step is itself a *universal* argument, not a check. That is what makes induction legitimate where sampling is not.

Section D — Challenge (optional)

6. Compute $5 + 9 + 13 + \cdots + (4n + 1)$ for $n = 1, 2, 3, 4$, conjecture a closed formula, and prove it by induction.

Approved solution

Compute: $n = 1$: 5. $n = 2$: 14. $n = 3$: 27. $n = 4$: 44.
These are $1 \cdot 5$, $2 \cdot 7$, $3 \cdot 9$, $4 \cdot 11$, suggesting the conjecture $n(2n + 3)$.
Proof. We proceed by induction. **Base step:** $n = 1$ gives $5 = 1 \cdot 5$. ✓
Inductive step. Assume $5+9+\cdots+(4k+1) = k(2k+3)$. The $(k+1)$st term is $4(k+1)+1 = 4k+5$. Then

$$k(2k + 3) + (4k + 5) = 2k^2 + 7k + 5 = (k + 1)(2k + 5) = (k + 1)\big(2(k + 1) + 3\big).$$

By induction the formula holds for every positive integer n. ■

Induction verifies a formula; it does not find one. The conjecture came from computing small cases and spotting a pattern — which is fine, because induction then supplies the proof those cases alone could not.

Instructor notes

- **Problem 4 is the key item** — a valid inductive step for a false claim, whose hypothesis is never true. It ties Lecture 4's vacuous truth to Lecture 28's warning.
- **Problem 5** is the conceptual payoff of the whole term. Students who can explain why the inductive step is universal have understood what induction is.
- **Problems 1–3** drill the "split off the last term" move. Problem 1's $(k+1)$st term is where most slips happen.
- **Problem 6** separates finding a formula from proving one — worth saying aloud.
- Reserved for Homework 11: Section 4.1 even-numbered exercises, including Exercise 2 (the telescoping sum) and Exercise 12, which is the same trap as Problem 4.